

Forbidden Comparisons in Fixed-Effect Poisson Difference-in-Differences

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Abstract

This paper studies the population treatment coefficient from pooling a staggered-adoption panel in a unit-and-time fixed-effect Poisson pseudo-likelihood. We show that this projection coefficient can be negative even when every cohort-time proportional treatment effect is positive. The failure reflects heterogeneous effects interacting with the fixed-effect PPML score: a four-cohort design with equal shares and flat untreated means provides an explicit sign reversal, while a positive-weight proportional treatment-on-the-treated target remains positive. The result characterizes the interpretation of the deterministic population projection targeted by the pooled criterion.

1 Introduction

Poisson pseudo-maximum likelihood with high-dimensional fixed effects is a standard way to estimate multiplicative mean models. It is especially common in gravity and other settings where the conditional mean is naturally proportional, zeros are empirically important, and fixed effects absorb large parts of the untreated outcome structure [Santos Silva and Tenreyro, 2006, 2011, Correia et al., 2020]. At the same time, many policy applications have staggered adoption: some units are treated early, others later, and some never. In linear difference-in-differences, it is now well understood that a pooled two-way fixed-effect coefficient can fail to aggregate heterogeneous effects with interpretable positive weights [Goodman-Bacon, 2021, de Chaisemartin and D’Haultfoeuille, 2020, Callaway and Sant’Anna, 2021, Sun and Abraham, 2021, Borusyak et al., 2024]. This paper studies the corresponding issue for a different object: the treatment coordinate of a pooled fixed-effect PPML population projection when treatment effects are proportional rather than additive.

The object of interest is deliberately narrow. Consider a finite cohort-time staggered-adoption design with positive limiting cohort shares, an exponential untreated mean with unit and time components, within-cohort baseline limits, proportional cohort-time treatment effects, and a full-rank collapsed fixed-effect design. The paper defines $\beta^*(\delta)$ in [definition 3](#) as the treatment coordinate of the limiting collapsed Poisson pseudo-likelihood projection. This is the population analogue of the coefficient an applied pooled unit-and-time fixed-effect PPML regression is trying to summarize. Its projection definition permits signed comparison weights and distinguishes it from causal average treatment effects built as convex aggregates of granular proportional effects.

The central question is whether the sign of this pooled PPML coefficient must agree with the signs of the underlying proportional effects. Sign reversal is possible. For any supported design

satisfying the horizon, no-first-period-adopter, rank, and maintained population restrictions and whose cohort support contains $\{2, 3, 4, \infty\}$, [theorem 3](#) gives a primitive sign index Φ . Its sign is exactly the sign of $\beta^*(\delta)$. The theorem’s closed-form polynomial calculation is a separate specialization to the exact four-cohort support $\{2, 3, 4, \infty\}$. The general sign equivalence also shows that $\Phi < 0$ implies

$$\beta^*(\delta) < 0 < PTT,$$

where PTT is the counterfactual-share proportional treatment-on-the-treated target. Thus the pooled fixed-effect PPML coefficient can have the opposite sign from every granular proportional effect while a positive-weight proportional target remains positive.

The mechanism is local as well as global. [Theorem 1](#) shows that, under collapsed rank, the derivative of $\beta^*(\delta)$ with respect to a treated cell’s log multiplier has the sign of the treatment indicator residual after projecting treatment on the fixed effects with weights $q_{gt}\mu_{gt}^*(\delta)$. These weights are generated by the fitted PPML mean through the pseudo-likelihood score and its curvature, producing a comparison geometry distinct from linear least squares. A treated cell with a negative weighted fixed-effect residual pushes the pooled coefficient downward when its proportional effect is increased. This is the multiplicative PPML analogue of a forbidden-comparison concern.

The paper also records the benchmark in which the pooled coefficient is well behaved. Under the same exponential untreated mean and collapsed rank conditions, if every treated cohort-time cell has the same log proportional effect δ_0 , [theorem 2](#) shows that $\beta^*(\delta) = \delta_0$ and that the fitted collapsed mean equals the true collapsed mean in every cell. This calibration locates the sign problem in the interaction between heterogeneous proportional effects and their compression into one pooled treatment coordinate in a staggered-adoption design.

The explicit witness is intentionally simple. [Definition 6](#) considers four periods and four cohorts, $\{2, 3, 4, \infty\}$, with equal cohort shares, flat untreated means, and strictly positive treated-cell log effects. All treated cells have multiplier 101/100 except cell $(2, 4)$, which has multiplier 4. [Theorem 4](#) shows that this primitive belongs to the sign-reversal region in [definition 5](#). The special four-cohort calculation in [theorem 3](#) gives

$$\Phi = \frac{5x^2 + 12x - 2y - 15}{16},$$

and the witness has $\Phi = -11559/32000$. Hence the negative coefficient occurs with equal shares, strictly positive effects, and flat untreated time components.

The distinction between the pooled coefficient and a positive proportional target is important for interpretation. Counterfactual-share proportional treatment-on-the-treated targets, such as those studied by [Moreau-Kastler \[2025\]](#), aggregate granular proportional effects using positive exposure-based weights. Under strictly positive granular effects, those targets preserve the positive sign. The pooled fixed-effect PPML coefficient studied here is different: it is a pseudo-true projection coefficient in the sense of misspecified likelihood theory [[White, 1982](#), [Gouriéroux et al., 1984b,a](#)]. Its sign is governed by the fixed-effect fit and the induced score equation, not by positive aggregation weights.

The contribution is a sharp population diagnostic. It characterizes the deterministic projection target, derives its local and global sign formulas, and proves that under stated staggered-adoption and multiplicative-mean restrictions its sign can oppose every underlying proportional effect. These results identify the interpretation problem that granular-effect estimation and sampling inference must address in subsequent work.

The rest of the paper proceeds as follows. Section 2 relates the result to linear and non-linear DiD, proportional treatment-on-the-treated targets, misspecified likelihood, and PPML applications. Section 3 defines the finite-array and collapsed population projections and the sign-reversal region. Section 4 gives the local derivative result, the homogeneous benchmark, the primitive sign frontier, and the four-cohort witness. Section 5 discusses interpretation and extensions. The appendix contains the collapse lemma, the pseudo-true projection lemma, proofs of the main results, and the verification note.

2 Related Literature

This paper contributes to the econometric literature on difference-in-differences by studying a particular population object: the treatment coordinate of a pooled fixed-effect Poisson pseudo-likelihood in a staggered-adoption design. The classical program-evaluation literature established difference-in-differences as a comparison of changes across treated and untreated groups [Ashenfelter and Card, 1985, Angrist and Pischke, 2009, Imbens and Wooldridge, 2009]. Semi-parametric and nonlinear extensions clarify how identifying restrictions and target parameters change beyond additive linear conditional-mean models [Abadie, 2005, Athey and Imbens, 2006, Sant’Anna and Zhao, 2020]. The present paper studies the projection target generated by a deliberately pooled multiplicative regression and asks when that common coefficient can have the wrong sign even though all cohort-time proportional effects are positive.

A large recent literature shows that two-way fixed-effect linear regressions in staggered-adoption designs can aggregate heterogeneous treatment effects through signed or hard-to-interpret weights. Goodman-Bacon [2021] decomposes the linear two-way fixed-effect coefficient into two-by-two comparisons and highlights the role of comparisons between already-treated and newly treated groups. de Chaisemartin and D’Haultfoeuille [2020], Callaway and Sant’Anna [2021], Sun and Abraham [2021], and Borusyak et al. [2024] develop alternative estimands and estimators that use interpretable comparisons under suitable identifying restrictions. Those results are primarily about additive treatment effects and linear least-squares projections. Here the regression is Poisson pseudo-maximum likelihood with unit and time fixed effects, and treatment effects are proportional. Pooling across cohorts and periods therefore operates through a different mechanism: the pseudo-true Poisson coefficient responds to the fixed-effect fit and the induced comparison weights in the multiplicative mean model.

The analysis also relates to nonlinear difference-in-differences. Wooldridge [2023] specifies nonlinear conditional-mean models whose parameters or average partial effects are given a causal interpretation under maintained identifying restrictions. Roth and Sant’Anna [2020] and Roth et al. [2023] emphasize how identifying assumptions, functional-form restrictions, and target parameters must be separated. The present paper holds the multiplicative untreated-mean restriction fixed and asks what sign is carried by the common treatment coordinate when heterogeneous cohort-time multipliers are pooled in a misspecified fixed-effect PPML criterion. Its object is the pseudo-true projection coefficient $\beta^*(\delta)$, which is distinct from a nonlinear-DiD effect parameter selected for causal interpretability.

The closest proportional-effect comparison is the counterfactual-share treatment-on-the-treated target studied by Moreau-Kastler [2025]. The objects differ in four steps. Its primitive effects are granular proportional effects; its target weights those effects by untreated counterfactual exposure; its sign is positive whenever every granular effect is positive; and its analysis is organized around identifying that causal aggregate. Here the same granular effects enter

observed cohort-time means, but the target is the treatment coordinate solving the pooled fixed-effect PPML score equations. Those score equations induce fitted-mean curvature weights and residualized comparisons in place of counterfactual shares. Consequently, the positive proportional PTT and the potentially sign-reversing $\beta^*(\delta)$ coexist as distinct summaries, and the sign-reversal result requires the pooled-score characterization developed here.

Methodologically, the paper builds on the theory of misspecified maximum likelihood and pseudo-likelihood. White [1982] formalizes pseudo-true parameters under misspecification, and Gouriéroux et al. [1984b,a] develop the pseudo-maximum-likelihood properties of Poisson estimators. Andrews [1988] provides general laws of large numbers for dependent and heterogeneous settings that underlie many asymptotic arguments. The formal results here characterize the deterministic population target and its causal interpretation, separating that question from convergence of an empirical estimator to its pseudo-true target.

Fixed-effect Poisson estimation is standard in applied microeconometrics and count-data models [Hausman et al., 1984, Cameron and Trivedi, 2013]. In international trade, PPML with high-dimensional fixed effects is a central empirical tool because it accommodates zeros and multiplicative gravity equations [Santos Silva and Tenreyro, 2006, 2011, Correia et al., 2020]. Gravity applications often combine multiplicative conditional means, rich fixed effects, and policy variation across country pairs or time [Anderson and van Wincoop, 2003, Head and Mayer, 2014, Yotov et al., 2016, Nagengast and Yotov, 2025]. Those features make PPML attractive, but they also make the pooled treatment coefficient easy to overinterpret when treatment timing is staggered and proportional effects vary across treated cells.

Recent work continues to refine fixed-effect and staggered-adoption estimators, including imputation and two-stage approaches for heterogeneous treatment effects [Gardner, 2022, Baker et al., 2025]. This paper complements that literature with a population characterization of when the coefficient produced by the familiar pooled fixed-effect PPML regression is negative despite strictly positive proportional effects. The result isolates the projection itself as the source of an unsafe causal summary and provides a diagnostic foundation for subsequent estimator and inference work.

3 Setup and Assumptions

Consider a finite- N panel with units $i = 1, \dots, N$ observed over calendar periods $t = 1, \dots, T$. Each unit belongs to an adoption cohort G_i in the finite support $\mathcal{C}$; the never-treated cohort is denoted by ∞ . Treatment is absorbing: D_{gt} indicates whether cohort g is treated in period t , and $D_{it} = D_{G_i t}$. Let $I_{gN} = \{i : G_i = g\}$ and $n_{gN} = |I_{gN}|$. Expectations under the triangular-array population law are written $E_{\mathcal{P}_N}$.

The first restrictions keep the large-array design stable and impose the multiplicative untreated mean structure. Potential outcomes are $Y_{it}(0)$ and $Y_{it}(1)$. The untreated mean has a positive unit baseline b_{iN} and a calendar component γ_{t0} ; $\bar{b}_{gN}$ is the within-cohort average baseline and $\bar{b}_g$ its limit.

Notation	Economic object
D_{gt}	treatment status of cohort g in period t
$B_{gt} = \bar{b}_g \exp(\gamma_{t0})$	untreated cohort-time mean
δ_{gt}	treated-cell log proportional effect
$m_{gt}(\delta) = B_{gt} \exp(D_{gt} \delta_{gt})$	observed cohort-time mean
$\beta_N^*(\delta)$	finite-array population PPML treatment coordinate
$\widehat{\beta}^*(\delta)$	limiting collapsed population PPML treatment coordinate
$\widehat{W}_{gt}(\delta)$	fitted-mean-weighted residualized treatment
$\Phi(\delta)$	primitive index with the sign of $\beta^*(\delta)$

Assumption 1. `[ass:cohort-share-limit]` (Cohort share limits) *For every $N \geq |\mathcal{C}|$ and every $g \in \mathcal{C}$,*

$$n_{gN} > 0 \quad \text{and} \quad \frac{n_{gN}}{N} \rightarrow \pi_g \in (0, 1).$$

[assumption 1](#) is the positive limiting cohort shares condition: every cohort remains represented asymptotically and contributes positive mass to the collapsed comparison [[de Chaisemartin and D’Haultfoeulle, 2020](#)]. The limit share is denoted π_g .

Assumption 2. `[ass:unit-untreated-exponential-mean]` (Untreated exponential mean) *For every unit $i = 1, \dots, N$ and period $t = 1, \dots, T$,*

$$E_{\mathcal{P}_N}[Y_{it}(0)] = b_{iN} \exp(\gamma_{t0}),$$

with the normalization $\gamma_{10} = 0$.

[assumption 2](#) is a unit-fixed-effect multiplicative parallel-trends restriction in an exponential mean model [[Wooldridge, 2023](#)]. The normalization fixes the location of the time effects and has no substantive content.

Assumption 3. `[ass:within-cohort-baseline-limit]` (Within-cohort baseline limits) *For every $g \in \mathcal{C}$, the deterministic triangular-array average*

$$\bar{b}_{gN} = n_{gN}^{-1} \sum_{i \in I_{gN}} b_{iN}$$

satisfies

$$\bar{b}_{gN} \rightarrow \bar{b}_g \in (0, \infty).$$

[assumption 3](#) is specific to this analysis. It is the condition that makes the cohort-time collapse deterministic: after averaging within a cohort, the only limiting untreated heterogeneity that remains is the cohort baseline $\bar{b}_g$.

Assumption 4. `[ass:proportional-effects]` (Proportional treatment effects) *For every unit $i = 1, \dots, N$ and period $t = 1, \dots, T$ with $D_{it} = 1$,*

$$E_{\mathcal{P}_N}[Y_{it}(1)] = E_{\mathcal{P}_N}[Y_{it}(0)] \exp(\delta_{G_{it}}).$$

[assumption 4](#) is the cohort-time proportional treatment effects condition [[Moreau-Kastler, 2025](#)]. The log multiplier $\delta_{G_{it}}$ may vary across treated cohorts and calendar periods, so the pooled coefficient below is not imposed to equal a common structural effect.

These assumptions induce a collapsed cohort-by-time representation. Define the untreated cohort-time mean $B_{gt} = \bar{b}_g \exp(\gamma_{t0})$, the collapsed cell mass $q_{gt} = \pi_g/T$, and the observed collapsed mean $m_{gt}(\delta) = B_{gt} \exp(D_{gt}\delta_{gt})$, where δ_{gt} denotes the cohort-time log multiplier on treated cells. Let r_{gt} collect the collapsed fixed-effect regressors and the treatment indicator, and let X_{gt} be the same vector with the treatment coordinate removed.

Assumption 5. [ass:collapsed-design-rank] (Collapsed design rank) *The collapsed second-moment matrix*

$$\sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} r_{gt} r'_{gt}$$

is positive definite.

[assumption 5](#) is the full-rank Poisson fixed-effect design condition [[White, 1982](#)]. It rules out exact collinearity in the collapsed regressors, so the population projection has a well-defined treatment coordinate rather than an unidentified linear combination of fixed effects and treatment.

The population object is a Poisson pseudo-likelihood projection defined through the conditional-mean criterion. For a finite array, let v_{itN} be the unit-time regressor vector containing unit effects, time effects, and treatment; let θ_N be the corresponding coefficient vector; and let $m_{itN}(\delta)$ be the observed unit-time mean generated by the untreated mean and proportional effects.

Definition 1. [def:unit-fe-population-projection] (Unit-FE Projection $\beta_N^*(\delta)$) *Define $\beta_N^*(\delta)$ as the last coordinate of a maximizer over $\theta_N \in \mathbb{R}^{N+T}$ of*

$$L_N(\theta_N; \delta) = (NT)^{-1} \sum_{i=1}^N \sum_{t=1}^T \{m_{itN}(\delta) v'_{itN} \theta_N - \exp(v'_{itN} \theta_N)\}.$$

[definition 1](#) is the finite-array population analogue of the unit fixed-effect PPML regression. The criterion $L_N(\theta_N; \delta)$ is evaluated at means rather than realized outcomes, and $\beta_N^*(\delta)$ is the treatment coordinate of that population projection.

Definition 2. [def:sample-unit-fe-ppml] (Sample PPML Coefficient $\hat{\beta}_N$) *Let A_N be the set of admissible extended maximizers of*

$$\sum_{i=1}^N \sum_{t=1}^T \{Y_{it} v'_{itN} \theta_N - \exp(v'_{itN} \theta_N)\}$$

over unit-effect coordinates in $\mathbb{R} \cup \{-\infty\}$, time-effect coordinates in $\mathbb{R}$ satisfying the period-1 normalization, and $\beta \in \mathbb{R}$, with the convention $\exp(-\infty) = 0$. Admissibility means that the period-1 time effect is normalized to zero and that a unit-effect coordinate may equal $-\infty$ only for a unit whose whole observed outcome path satisfies $Y_{it} = 0$ for every t ; for such a unit the extended objective contribution is defined to be zero. If A_N contains an element whose squared Euclidean norm over the finite time-effect and β coordinates is no larger than that of every element of A_N , then $\hat{\beta}_N$ is the β -coordinate of such a selected element. If no such minimum-finite-coordinate-norm admissible extended maximizer exists, set $\hat{\beta}_N = 0$.

The limiting projection works directly with the collapsed cohort-time cells. With θ denoting the collapsed coefficient vector, write the limiting collapsed Poisson criterion as $L(\theta; \delta)$. The finite collapsed analogue, used only to connect the array to this limit, is denoted $L_{Nc}(\theta; \delta)$.

Definition 3. [def:collapsed-population-projection] (Collapsed Projection $\theta^*(\delta)$) *Define*

$$\theta^*(\delta) \in \arg \max_{\theta \in \mathbb{R}^{|C|+T}} L(\theta; \delta),$$

where

$$L(\theta; \delta) = \sum_{g \in C} \sum_{t=1}^T q_{gt} \{m_{gt}(\delta) r'_{gt} \theta - \exp(r'_{gt} \theta)\}.$$

The coefficient $\beta^*(\delta)$ is the last coordinate of $\theta^*(\delta)$, and the fitted collapsed mean is

$$\mu_{gt}^*(\delta) = \exp\{r'_{gt} \theta^*(\delta)\}.$$

[definition 3](#) gives the main object studied in the paper. The maximizer $\theta^*(\delta)$, its treatment coordinate $\beta^*(\delta)$, and the fitted mean $\mu_{gt}^*(\delta)$ are pseudo-true quantities in the sense of misspecified likelihood theory [[White, 1982](#), [Gouriéroux et al., 1984b,a](#)]. The target is therefore a deterministic population projection defined by the Poisson mean criterion.

The analysis centers on this deterministic population projection. A sample fixed-effect PPML regression replaces the means by observed outcomes and may also confront separation [[Santos Silva and Tenreiro, 2006, 2011](#), [Correia et al., 2020](#)]; its sampling theory is a distinct extension. Accordingly, references below to a “pooled coefficient” mean the population coordinate $\beta^*(\delta)$, unless explicitly stated otherwise.

The sign analysis uses the residualized treatment after partialling out the collapsed fixed effects with weights given by the fitted PPML mean. Let $\rho^*(\delta)$ be the weighted least-squares coefficient from that auxiliary projection.

Definition 4. [def:weighted-fwl-residual] (Weighted Residual $\widetilde{W}_{gt}(\delta)$) *Define*

$$\rho^*(\delta) \in \arg \min_{\rho \in \mathbb{R}^{|C|+T-1}} \sum_{g \in C} \sum_{t=1}^T q_{gt} \mu_{gt}^*(\delta) \{D_{gt} - X'_{gt} \rho\}^2.$$

The weighted Frisch–Waugh–Lovell residualized treatment is

$$\widetilde{W}_{gt}(\delta) = D_{gt} - X'_{gt} \rho^*(\delta).$$

[definition 4](#) gives the comparison weights that enter the local sign formula in the next section. The residual $\widetilde{W}_{gt}(\delta)$ is the fixed-effect projection residual of treatment under the curvature weights of the Poisson criterion and serves as a projection comparison weight.

The sign-reversal result is stated on designs with at least the four adoption statuses needed for the explicit construction. The next two assumptions isolate that scope and the positive-effect case of interest.

Assumption 6. [ass:multicohort-frontier-scope] (Multicohort frontier scope) *The cohort set satisfies*

$$\{2, 3, 4, \infty\} \subseteq C.$$

[assumption 6](#) is specific to this analysis. It ensures that the design contains early, middle, late, and never-treated cohorts, the minimal structure used by the four-cohort construction.

Assumption 7. `[ass:strict-positive-effects]` (Strict Positive Effects) *For every cohort-period cell (g, t) with $D_{gt} = 1$,*

$$\delta_{gt} > 0.$$

[assumption 7](#) is also specific to this analysis. It deliberately restricts attention to the strongest sign benchmark: every treated cell has a positive proportional effect, so any negative pooled coefficient is a property of the projection rather than a mixture of positive and negative underlying effects.

The global parameter region collects the maintained restrictions under which the limiting PPML treatment coordinate is negative despite those positive cell-level effects.

Definition 5. `[def:global-sign-reversal-region]` (Sign-Reversal Region $\mathcal{R}_T$) *Define $\mathcal{R}_T$ as the set of all triples*

$$((\pi_g)_{g \in \mathcal{C}}, (B_{gt})_{g \in \mathcal{C}, 1 \leq t \leq T}, (\delta_{gt})_{(g,t): D_{gt}=1})$$

with $T \geq 4$, indexed by a cohort support $\mathcal{C}$ that contains $\{2, 3, 4, \infty\}$ and contains no first-period finite cohort, such that

$$0 < \pi_g < 1 \quad (g \in \mathcal{C}), \quad \sum_{g \in \mathcal{C}} \pi_g = 1,$$

there exist $\bar{b}_g > 0$ for $g \in \mathcal{C}$ with $B_{gt} = \bar{b}_g \exp(\gamma_{t0})$ for every supported cell (g, t) , $\delta_{gt} > 0$ for every treated supported cell (g, t) with $D_{gt} = 1$, the collapsed design has full weighted rank in the sense that

$$\forall a \neq 0, \quad 0 < \sum_{g \in \mathcal{C}} \sum_{t=1}^T \frac{\pi_g}{T} (r'_{gt} a)^2,$$

and

$$\beta^*(\delta) < 0.$$

[definition 5](#) names the population configurations in which the pooled limiting coefficient has the opposite sign from every granular proportional treatment effect. The notation $\mathcal{R}_T$ is only a shorthand for this primitive set of shares, untreated means, and treated-cell multipliers.

The paper uses one explicit triangular array to show that this region is nonempty. Its role is evidentiary rather than calibrational: all baselines and untreated time effects are set to one, and the only large treated multiplier is placed in the period-4 cell of the cohort first treated in period 2.

Definition 6. `[def:four-cohort-witness]` (Four-cohort witness W_4) *Let W_4 be the cofinal triangular subsequence indexed by $N = 4m$, $m = 1, 2, \dots$, with $T = 4$ and*

$$\mathcal{C} = \{2, 3, 4, \infty\}.$$

For each such N ,

$$n_{2N} = n_{3N} = n_{4N} = n_{\infty N} = m,$$

$b_{iN} = 1$ for every i , and $\gamma_{t0} = 0$ for every t . The treatment effects satisfy

$$\delta_{gt} = \log(101/100)$$

for every treated cell $(g, t) \neq (2, 4)$, while

$$\delta_{2,4} = \log(4).$$

[definition 6](#) defines the witness W_4 . Equal cohort shares and flat untreated means remove scale and composition as explanations for the sign reversal; the later result depends only on the staggered treatment pattern and the heterogeneous positive proportional effects encoded here.

Finally, the primitive sign characterization uses an algebraic index based on observed mean components. Let H denote the set of treated cohort-time cells when that notation is needed later.

Definition 7. `[def:frontier-elimination-handle]` (Frontier-elimination polynomial $\Phi(\delta)$)
Define

$$\Phi(\delta) = MA - \sum_{g,t} D_{gt} R_g C_t,$$

where

$$h_{gt} = \pi_g B_{gt} \exp(D_{gt} \delta_{gt}), \quad R_g = \sum_t h_{gt}, \quad C_t = \sum_g h_{gt},$$

and

$$M = \sum_g R_g, \quad A = \sum_{g,t} D_{gt} h_{gt}.$$

The relation between $\Phi(\delta)$ and $\beta^*(\delta)$ is stated separately.

[definition 7](#) introduces the scalar $\Phi(\delta)$ used by the primitive sign result. The component h_{gt} is the cohort-share-weighted observed collapsed mean, equivalently $Tq_{gt}m_{gt}(\delta)$, while R_g , C_t , M , and A are its cohort sums, period sums, grand sum, and treated-cell sum. The main theorem in the next section supplies the sign relation for this scalar.

4 Main Results

The setup isolates a population object: the treatment coordinate of the collapsed fixed-effect PPML projection in [definition 3](#). The first result gives the local comparative static of that coefficient with respect to one treated cohort-time log multiplier. Its content is a nonlinear analogue of a forbidden-comparison diagnostic: after the Poisson fixed effects have been fit, the sign of the derivative is exactly the sign of the residualized treatment in that cell, where residualization uses the fitted PPML mean as the curvature weight.

Theorem 1. `[thm:sharp-ppml-forbidden-sign]` (Sharp PPML Sign) *Let T be positive and let $\mathcal{C}$ be a nonempty finite cohort set. Fix cohort shares $\pi_g \in (0, 1)$, baseline levels $\bar{b}_g > 0$, time effects γ_t , and a proportional-effect array δ . Assume:*

- **(Rank.)** *The collapsed design satisfies [assumption 5](#).*
- **(Cell.)** *The cohort k belongs to $\mathcal{C}$, the period $s \in \{1, \dots, T\}$, and the cohort-period cell is treated: $D_{ks} = 1$.*

Write

$$\widetilde{W}_{ks}(\delta)$$

for the pseudo-true $q_{gt}\mu_{gt}^*(\delta)$ -weighted FWL residual of the treatment indicator in cell (k, s) , and define

$$\mathcal{E}(\delta) = \sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} \mu_{gt}^*(\delta) \widetilde{W}_{gt}(\delta)^2.$$

Then $\mathcal{E}(\delta) > 0$, and the one-coordinate path

$$x \mapsto \beta^*(\delta \text{ with } \delta_{ks} \text{ replaced by } x)$$

is differentiable at $x = \delta_{ks}$ with derivative

$$\frac{q_{ks} B_{ks} \exp(\delta_{ks}) \widetilde{W}_{ks}(\delta)}{\mathcal{E}(\delta)}.$$

Consequently,

$$\frac{\partial \beta^*(\delta)}{\partial \delta_{ks}} < 0 \iff \widetilde{W}_{ks}(\delta) < 0, \quad \frac{\partial \beta^*(\delta)}{\partial \delta_{ks}} = 0 \iff \widetilde{W}_{ks}(\delta) = 0,$$

and

$$\frac{\partial \beta^*(\delta)}{\partial \delta_{ks}} > 0 \iff \widetilde{W}_{ks}(\delta) > 0.$$

[assumption 5](#) makes the denominator $\mathcal{E}(\delta)$ strictly positive by preserving residual treatment variation after projecting on the fixed effects. Thus the local sign follows directly from the pseudo-true fit. A treated cell with a negative residual in [definition 4](#) pushes the pooled coefficient downward when its proportional effect is increased; a positive residual pushes it upward. This is the PPML counterpart to the comparison-weight concern in staggered-adoption linear regressions [[Goodman-Bacon, 2021](#), [de Chaisemartin and D’Haultfoeuille, 2020](#)], with weights generated by the multiplicative pseudo-likelihood projection instead of least squares on the outcome.

The next result records the exact-calibration benchmark. When all treated cells have the same proportional log effect, the fixed-effect PPML projection recovers that common multiplier exactly.

Theorem 2. [`prop:homogeneous-effect-reduction`] (Homogeneous Effect Reduction) *Fix T , a finite cohort set $\mathcal{C}$, sampling laws $\{\mathcal{P}_N\}_N$, potential outcomes $Y_{it}(d)$, positive unit baselines b_{iN} , cohort shares $\pi_g \in (0, 1)$, positive limiting cohort baselines $\bar{b}_g$, untreated time components γ_{t0} , cohort-time log effects δ_{gt} , and a scalar δ_0 . Suppose that*

- *(Untreated means.)* [assumption 2](#) holds.
- *(Collapsed rank.)* [assumption 5](#) holds for $\mathcal{C}$ and $\{\pi_g\}_g$.
- *(Common treated-cell effect.)* For every $g \in \mathcal{C}$ and every period t , if $D_{gt} = 1$, then $\delta_{gt} = \delta_0$.

Then

$$\beta^*(\delta) = \delta_0,$$

and, for every $g \in \mathcal{C}$ and every period t ,

$$\mu_{gt}^*(\delta) = B_{gt} \exp(D_{gt} \delta_0).$$

The homogeneous case is therefore an exact calibration point. If the proportional effect is common across treated cells, the treatment regressor completes the multiplicative mean specification after cohort and time effects are included, and the pooled coefficient has the expected

causal interpretation. Heterogeneous proportional effects generate the sign-reversal mechanism studied here, the setting in which modern staggered-adoption estimands use cohort-specific effects and interpretable aggregation [Callaway and Sant’Anna, 2021, Sun and Abraham, 2021, Borusyak et al., 2024].

The sharp local derivative identifies the cellwise mechanism. The next theorem complements it with a primitive scalar index whose sign is exactly the sign of the limiting pooled PPML coefficient under the maintained multicohort positive-effect scope. It also compares the pooled coefficient with the counterfactual-share proportional treatment-on-the-treated target, which remains positive when every granular proportional effect is positive [Moreau-Kastler, 2025].

Theorem 3. [thm:primitive-global-frontier] (Primitive Sign Frontier) *Fix a panel length T , a finite cohort support $\mathcal{C}$, triangular-array sampling laws $\mathcal{P}_N$, potential outcomes $Y_{it}(d)$, deterministic cohort labels G_i , positive unit baselines b_{iN} , limiting shares $\pi_g \in (0, 1)$, positive limiting cohort baselines $\bar{b}_g$, untreated time effects γ_{t0} , and log proportional effects δ_{gt} . Suppose:*

- **(Horizon and support.)** $T > 0$, $T \geq 4$, the never-treated cohort belongs to $\mathcal{C}$, and every finite supported cohort has adoption date different from the first period.
- **(Array support.)** For every N and every unit i , $G_i \in \mathcal{C}$.
- **(Share limits.)** [assumption 1](#) holds for $(\mathcal{C}, G_i, \pi_g)$.
- **(Untreated means.)** [assumption 2](#) holds for $(\mathcal{P}_N, Y_{it}(d), b_{iN}, \gamma_{t0})$.
- **(Baseline limits.)** [assumption 3](#) holds for $(\mathcal{C}, G_i, b_{iN}, \bar{b}_g)$.
- **(Proportional effects.)** [assumption 4](#) holds for $(\mathcal{P}_N, Y_{it}(d), G_i, \delta_{gt})$.
- **(Rank.)** [assumption 5](#) holds for $(\mathcal{C}, \pi_g)$.
- **(Frontier scope.)** [assumption 6](#) holds for $(T, \mathcal{C})$.
- **(Positive effects.)** [assumption 7](#) holds for $(\mathcal{C}, \delta_{gt})$.

Define

$$h_{gt} = \pi_g \bar{b}_g \exp(\gamma_{t0}) \exp(D_{gt} \delta_{gt}), \quad R_g = \sum_{t=1}^T h_{gt}, \quad C_t = \sum_{g \in \mathcal{C}} h_{gt},$$

$$M = \sum_{g \in \mathcal{C}} R_g, \quad A = \sum_{g \in \mathcal{C}} \sum_{t=1}^T D_{gt} h_{gt}, \quad \Phi = MA - \sum_{g \in \mathcal{C}} \sum_{t=1}^T D_{gt} R_g C_t.$$

Then

$$\beta^*(\delta) < 0 \iff \Phi < 0, \quad \beta^*(\delta) = 0 \iff \Phi = 0, \quad 0 < \beta^*(\delta) \iff 0 < \Phi.$$

Moreover, the primitive tuple induced by $(\pi_g, \bar{b}_g, \gamma_{t0}, \delta_{gt})$ belongs to $\mathcal{R}_T$ if and only if $\Phi < 0$.

For every $x > 1$ and $y > 1$, in the four-period support $\{2, 3, 4, \infty\}$ with equal shares, unit limiting baselines, zero untreated time effects, multiplier x on treated cells other than $(2, 4)$, and multiplier y at $(2, 4)$,

$$\Phi = \frac{5x^2 + 12x - 2y - 15}{16}.$$

Also,

$$\frac{5(101/100)^2 + 12(101/100) - 15}{2} = \frac{4441}{4000},$$

and the explicit witness W_4 has

$$\Phi = -\frac{11559}{32000}.$$

Let

$$m_{gt} = \bar{b}_g \exp(\gamma_{t0}) \exp(D_{gt}\delta_{gt}), \quad H = \{(g, t) : g \in \mathcal{C}, D_{gt} = 1\}.$$

For every $(g, t) \in H$,

$$B_{gt}^{obs} = B_{gt}, \quad \tau_{gt} = \exp(\delta_{gt}) - 1, \quad \omega_{gt} > 0.$$

The counterfactual-share weights sum to one:

$$\sum_{(g,t) \in H} \omega_{gt} = 1.$$

The counterfactual-share PTT satisfies

$$PTT = \frac{\sum_{(g,t) \in H} q_{gt} m_{gt}}{\sum_{(g,t) \in H} q_{gt} B_{gt}^{obs}} - 1.$$

Finally,

$$\Phi < 0 \implies \beta^*(\delta) < 0 < PTT.$$

[theorem 3](#) has two distinct scopes. The sign equivalence between Φ and the pooled coefficient applies to every support satisfying the theorem's assumptions, including supports that strictly contain $\{2, 3, 4, \infty\}$. The displayed polynomial in x and y , and the numerical value for W_4 , specialize that general characterization to the exact equal-share four-cohort support. The index Φ serves as a population diagnostic conditional on known or specified primitives, while the counterfactual-share proportional PTT is the causal aggregate. In the same primitive configurations, the counterfactual-share weights are positive and sum to one, so the proportional PTT is a positive weighted average whenever all granular proportional effects are positive.

The explicit four-cohort construction makes the sign reversal concrete. The support contains cohorts first treated in periods 2, 3, and 4, together with a never-treated cohort; untreated means are flat and cohort shares are equal. Hence the example establishes the phenomenon under conventional baseline scaling and constant untreated time components.

Theorem 4. `[prop:four-cohort-sign-reversal]` (Four-Cohort Sign Reversal) *The four-cohort witness W_4 is constructed with cohort support $\mathcal{C} = \{2, 3, 4, \infty\}$, equal shares $\pi_g = 1/4$, limiting baselines $\bar{b}_g = 1$, untreated time component $\gamma_{t0} = 0$, and proportional-effect log multipliers*

$$\delta_{gt} = \begin{cases} \log 4, & (g, t) = (2, 4), \\ \log(101/100), & D_{gt} = 1 \text{ and } (g, t) \neq (2, 4), \\ 0, & D_{gt} = 0. \end{cases}$$

Then:

- **(Region membership.)** The primitive W_4 belongs to the sign-reversal region $\mathcal{R}_4$ of [definition 5](#).
- **(Positive effects.)** Every treated supported cell has a strictly positive proportional-effect log multiplier: if $g \in \mathcal{C}$, $t \in \{1, 2, 3, 4\}$, and $D_{gt} = 1$, then $\delta_{gt} > 0$.
- **(Unique largest treated effect.)** The treated cell $(2, 4)$ has the strictly largest proportional-effect log multiplier: if $g \in \mathcal{C}$, $t \in \{1, 2, 3, 4\}$, $D_{gt} = 1$, and $(g, t) \neq (2, 4)$, then $\delta_{gt} < \delta_{2,4}$.
- **(Weighted FWL residuals.)** At the no-effect vector, the weighted FWL residual $\widetilde{W}_{gt}$ of [definition 4](#) equals

$$\frac{1}{8} \begin{pmatrix} -3 & 3 & 1 & -1 \\ -1 & -3 & 3 & 1 \\ 1 & -1 & -3 & 3 \\ 3 & 1 & -1 & -3 \end{pmatrix},$$

with rows $g = 2, 3, 4, \infty$ and columns $t = 1, 2, 3, 4$. In particular, $\widetilde{W}_{2,4} = -1/8$.

- **(Negative late-cell derivative at zero.)** Holding all other effects at zero,

$$\left. \frac{d}{dx} \beta^*(\delta_{2,4} = x, \delta_{gt} = 0 \text{ for } (g, t) \neq (2, 4)) \right|_{x=0} = -\frac{1}{10}.$$

- **(Local negative derivative under positive effects.)** There exists $\varepsilon > 0$ such that, for every effect vector δ' satisfying $|\delta'_{gt}| < \varepsilon$ for all $g \in \mathcal{C}$ and $t \in \{1, 2, 3, 4\}$, if δ' has strictly positive treated effects in the sense of [assumption 7](#), then

$$\left. \frac{d}{dx} \beta^*(\delta' \text{ with } \delta'_{2,4} \text{ replaced by } x) \right|_{x=\delta'_{2,4}} < 0.$$

The residual table in [theorem 4](#) identifies the period-4 cell $(2, 4)$ of the early-treated cohort as locally negative at the no-effect vector, and the neighborhood statement extends that derivative sign to small strictly positive effects. The actual witness W_4 , however, is a global sign-reversal example because [theorem 3](#) gives $\Phi < 0$ for its primitive tuple and [theorem 4](#) places that tuple in $\mathcal{R}_4$. In this example every treated cell has a positive proportional effect, and the largest effect is in that late calendar cell, yet the limiting pooled PPML coefficient is negative.

The main implication concerns interpretation under heterogeneous proportional effects: the single pooled coefficient is a pseudo-true projection whose sign is governed by fixed-effect residual comparisons. Interpretable causal summaries should therefore be built from granular proportional effects and positive aggregation weights, as in counterfactual-share targets, in place of inference from the sign of the pooled fixed-effect PPML coefficient alone [[Moreau-Kastler, 2025](#), [Wooldridge, 2023](#), [Roth and Sant'Anna, 2020](#)].

5 Discussion and Extensions

The results matter most in empirical settings where the multiplicative conditional mean is substantively natural and the treatment timing is staggered. Gravity applications are a leading example: PPML specifications with high-dimensional fixed effects are standard because trade flows

are modeled through multiplicative resistance terms, exposure variables, and policy shifters, and because zeros are common in the outcome [Anderson and van Wincoop, 2003, Head and Mayer, 2014, Yotov et al., 2016, Santos Silva and Tenreyro, 2006, 2011, Correia et al., 2020]. In such designs, a pooled treatment indicator may look like a compact summary of a policy change. The results above show that, under heterogeneous proportional effects, the population coefficient of that regression is instead a fixed-effect Poisson projection whose sign can be governed by residualized comparisons across cohort-time cells.

The conclusion pinpoints how PPML summarizes a multiplicative mean model. The homogeneous benchmark in [theorem 2](#) shows that a common proportional effect is recovered exactly. With heterogeneous cohort-time proportional effects, [theorem 1](#) shows that increasing a positive treated-cell effect moves the pooled coefficient in the direction determined by the weighted fixed-effect residual for that cell. The interpretation problem therefore arises from compressing a collection of heterogeneous economic effects into a single pooled projection coordinate.

5.1 Limitations and future work

The primitive index Φ in [theorem 3](#) is a population diagnostic conditional on known or specified primitives. It determines whether the limiting fixed-effect PPML treatment coordinate is negative, zero, or positive. Estimation of those primitives, implementation of Φ as a specification test, and construction of positive-weight estimators from granular proportional effects are the main next steps.

That distinction is important for empirical interpretation. A negative pooled PPML coefficient is sometimes read as evidence that the treatment reduced the conditional mean. Under the conditions of [theorem 3](#), the same primitive configuration can instead have $\beta^*(\delta) < 0$ and a positive counterfactual-share proportional treatment-on-the-treated target. The sign reversal therefore identifies the pooled fixed-effect PPML coefficient as a projection summary distinct from positive-weight proportional targets.

The positive counterfactual-share target remains closer to the causal object under the maintained proportional-effect restrictions because it aggregates treated-cell effects with weights tied to untreated counterfactual exposure [Moreau-Kastler, 2025]. When each granular proportional effect is positive, such an aggregation preserves the sign by construction. [Theorem 3](#) formalizes this contrast in the same population environment: the pooled coefficient can be negative while PTT is positive. Empirical conclusions should therefore distinguish the regression projection from the proportional treatment-on-the-treated estimand.

The four-cohort witness in [theorem 4](#) establishes the sign reversal with equal cohort shares, flat untreated time effects, and positive treated-cell effects. Staggered timing and heterogeneous proportional effects place a large positive effect in a cell with a negative residualized treatment value. This makes the phenomenon relevant for conventional staggered-adoption designs, including policy settings where treatment intensity or exposure plausibly varies across treated cohorts and periods.

The population characterization also clarifies the choice that precedes estimation. A researcher may intend a common log multiplier, a collection of granular proportional effects, or a positive-weight proportional aggregate; the pooled projection equals the first under homogeneity and can differ sharply from the third under heterogeneity. The primitive index Φ establishes this distinction theoretically. Future work can turn that diagnostic into an operational procedure by estimating granular effects, constructing a sampling distribution for Φ , assessing separation, and determining how often the sign-reversal region occurs in empirically calibrated designs [San-

tos Silva and Tenreiro, 2006, 2011, Correia et al., 2020, Wooldridge, 2023, Nagengast and Yotov, 2025].

6 Appendix: Proofs, Auxiliary Lemmas, and Verification Note

This appendix records the two auxiliary population results used to connect the unit-level fixed-effect regression to the collapsed cohort-time analysis. The ordering mirrors the logic of the main text. The collapse result first justifies replacing the finite- N unit fixed-effect projection by its cohort-time counterpart. The projection result then gives the first-order conditions for the limiting collapsed pseudo-true parameter. The main theorem proofs use these statements before establishing the primitive sign characterization in [theorem 3](#); the four-cohort sign-reversal result in [theorem 4](#) is then read as a consequence of that characterization together with the explicit witness construction.

The first lemma formalizes the reduction from a unit fixed-effect PPML population criterion to a collapsed criterion. Its role is to show that, under the same cohort-share, untreated-mean, baseline-limit, proportional-effect, and rank restrictions used in [assumption 1](#), the treatment coordinate is invariant to within-cohort rearrangements of unit baselines that leave cohort averages unchanged. Thus the finite-array object in [definition 1](#) and the limiting object in [definition 3](#) are linked through the cohort-time means that drive the sign analysis.

Lemma 1. [[lem:unit-fe-collapse](#)] (Unit FE Collapse) *Let T be a number of periods and let $\mathcal{C}$ be a finite cohort set. Suppose:*

- (*Horizon.*) $T \geq 4$.
- (*Cohort support.*) *The never-treated cohort belongs to $\mathcal{C}$, and every finite cohort in $\mathcal{C}$ is dated after the first period.*
- (*Array support.*) *For every N and every unit i , the cohort label G_i belongs to $\mathcal{C}$.*
- (*Positive primitives.*) *The unit baselines b_{iN} and the limiting within-cohort baselines $\bar{b}_g$ are strictly positive, and each limiting cohort share satisfies $\pi_g \in (0, 1)$.*
- (*Share limits.*) *[assumption 1](#) holds for G_i and π_g .*
- (*Untreated means.*) *[assumption 2](#) holds for $Y_{it}(d)$, $\mathcal{P}_N$, b_{iN} , and γ_{t0} .*
- (*Baseline limits.*) *[assumption 3](#) holds for G_i , b_{iN} , and $\bar{b}_g$.*
- (*Proportional effects.*) *[assumption 4](#) holds for $Y_{it}(d)$, $\mathcal{P}_N$, G_i , and δ_{gt} .*
- (*Collapsed rank.*) *[assumption 5](#) holds for $\mathcal{C}$ and π_g .*

Then, for every N with $|\mathcal{C}| \leq N$, the finite-array unit-FE population criterion $L_N(\theta_N; \delta)$ of [definition 1](#) has a unique global maximizer, the collapsed finite-array criterion $L_{Nc}(\theta; \delta)$ has a unique global maximizer, and

$$\beta_N^*(\delta) = \text{the treatment coordinate of the unique maximizer of } L_{Nc}(\theta; \delta).$$

Moreover,

$$\beta_N^*(\delta) \longrightarrow \beta^*(\delta) \quad \text{as } N \rightarrow \infty.$$

Finally, for every N with $|\mathcal{C}| \leq N$ and every alternative positive baseline array b'_{iN} , if

$$\bar{b}'_{gN} = \bar{b}_{gN} \quad \text{for every } g \in \mathcal{C},$$

then replacing b_{iN} by b'_{iN} leaves the finite-array unit-FE coefficient unchanged:

$$\beta_N^*(\delta; b') = \beta_N^*(\delta; b).$$

The uniqueness clauses are population pseudo-likelihood statements in the usual misspecified-likelihood sense [White, 1982, Gouriéroux et al., 1984b,a]. The final invariance clause is specific to the cohort-time collapse used here: once unit effects absorb individual baseline levels, only the within-cohort baseline average enters the collapsed mean relevant for the treatment coordinate. This is why the later sign results can be stated in terms of cohort shares, untreated cohort-time means, and proportional-effect multipliers rather than the full list of unit baselines.

The second auxiliary lemma records the first-order conditions for the collapsed pseudo-true PPML projection. These score equations are the fixed-effect and treatment equations behind the derivative formula in theorem 1 and the algebraic sign calculation in theorem 3.

Lemma 2. [lem:pseudo-true-ppml-projection] (Unique PPML Projection) *Fix a number of periods T , a finite cohort set $\mathcal{C}$, cohort shares $\pi_g \in (0, 1)$, positive limiting baseline means $\bar{b}_g > 0$, time components γ_{t0} , and proportional-effect log multipliers δ_{gt} . Suppose the collapsed design rank condition in assumption 5 holds for $(T, \mathcal{C}, \pi)$. Let $\theta^*(\delta)$, $L(\theta; \delta)$, $m_{gt}(\delta)$, and $\mu_{gt}^*(\delta)$ be as in definition 3. Then $\theta^*(\delta)$ is the unique global maximizer of $L(\theta; \delta)$. Moreover, for every collapsed nuisance regressor index j ,*

$$\sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} X_{gt,j} \{m_{gt}(\delta) - \mu_{gt}^*(\delta)\} = 0,$$

and the treatment score also vanishes:

$$\sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} D_{gt} \{m_{gt}(\delta) - \mu_{gt}^*(\delta)\} = 0.$$

lemma 2 is the collapsed analogue of the pseudo-true score condition for a Poisson quasi-likelihood projection. The rank condition rules out flat directions in the fixed-effect and treatment regressors, while positivity of the cell means keeps the exponential criterion strictly concave on the relevant design span. The argument uses the mean criterion characteristic of standard pseudo-maximum-likelihood analysis [White, 1982, Gouriéroux et al., 1984a].

Combining lemma 1 and lemma 2 fixes the population target before the sign results are applied. The derivative result in theorem 1 uses the score equations to express the response of the treatment coordinate through the weighted residualized treatment. The homogeneous benchmark in theorem 2 then identifies the correctly specified common-effect case. The primitive sign theorem, theorem 3, comes next because it supplies the global sign criterion used to place the explicit four-cohort construction in $\mathcal{R}_4$. With that ordering, theorem 4 becomes the concrete specialization of the sign frontier to the witness in definition 6.

Reproducibility note. The formal supplement verifies the displayed population claims in Lean 4 and records the correspondence between mathematical statements and declarations. Its deterministic scope covers the finite-dimensional projection, collapse, derivative-sign, primitive-sign, and four-cohort calculations. Sampling consistency, inference, and empirical identification of the primitive means and proportional effects form the natural empirical extension.

7 Proofs of the main results

Proof of [theorem 1](#). Write $W_{gt} = \widetilde{W}_{gt}(\delta)$, $\mu_{gt}^* = \mu_{gt}^*(\delta)$, and

$$\mathcal{E} = \sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} \mu_{gt}^* W_{gt}^2.$$

A collapsed parameter is written $\theta = (a, \beta)$ with $r'_{gt}\theta = \sum_j X_{gt,j} a_j + D_{gt}\beta$, and $\rho^*(\delta)$ is the weighted nuisance coefficient of [definition 4](#), so that

$$W_{gt} = D_{gt} - \sum_j X_{gt,j} \rho_j^*(\delta).$$

Step 1: $\mathcal{E} > 0$. Each summand is nonnegative, since $q_{gt} > 0$, $\mu_{gt}^* = \exp\{r'_{gt}\theta^*(\delta)\} > 0$, and $W_{gt}^2 \geq 0$. If $\mathcal{E} = 0$, every summand vanishes, and positivity of $q_{gt}\mu_{gt}^*$ forces $W_{gt} = 0$ at every supported cell. Consider the collapsed parameter

$$a = (-\rho^*(\delta), 1),$$

whose treatment coordinate is one, so $a \neq 0$. Its collapsed index is exactly the residual,

$$r'_{gt}a = D_{gt} - \sum_j X_{gt,j} \rho_j^*(\delta) = W_{gt} = 0 \quad \text{at every supported cell,}$$

whence $\sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} (r'_{gt}a)^2 = 0$, contradicting [assumption 5](#). Thus $\mathcal{E} > 0$.

Step 2: the perturbed criterion. For $x \in \mathbb{R}$ let δ^x be δ with the coordinate δ_{ks} replaced by x , and write $x_0 = \delta_{ks}$. Because $D_{ks} = 1$ and $m_{gt}(\delta) = B_{gt} \exp(D_{gt}\delta_{gt})$, the perturbation moves exactly one collapsed mean:

$$m_{ks}(\delta^x) = B_{ks} \exp(x), \quad m_{gt}(\delta^x) = m_{gt}(\delta) \quad \text{for } (g, t) \neq (k, s),$$

Setting

$$p_{gt} = \begin{cases} B_{ks} \exp(\delta_{ks}), & (g, t) = (k, s), \\ 0, & (g, t) \neq (k, s), \end{cases} \quad \text{we have} \quad \left. \frac{d}{dx} m_{gt}(\delta^x) \right|_{x=x_0} = p_{gt}.$$

Step 3: the score system and its solution branch. For a direction d put

$$S_d(x, \theta) = \sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} (r'_{gt}d) \{m_{gt}(\delta^x) - \exp(r'_{gt}\theta)\}.$$

By [lemma 2](#) applied at the effect array δ^x , the criterion $L(\cdot; \delta^x)$ has a unique global maximizer, and $S_d(x, \theta) = 0$ for every d both holds at that maximizer and, conversely, characterizes it: vanishing of every directional score is sufficient for global maximality of a Poisson criterion with nonnegative weights, by the tangent-line bound $\exp(\eta') \geq \exp(\eta)\{1 + (\eta' - \eta)\}$. The system is smooth in (x, θ) , it is solved at $(x_0, \theta^*(\delta))$, and its derivative in θ at that point is the bilinear form

$$d, e \mapsto - \sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} \mu_{gt}^* (r'_{gt}d)(r'_{gt}e),$$

which is negative definite — $e \neq 0$ gives some supported cell with $r'_{gt}e \neq 0$ by the injectivity of $\theta \mapsto (r'_{gt}\theta)_{(g,t)}$ established from [assumption 5](#) in [lemma 2](#) — and hence invertible. The implicit function theorem therefore supplies a continuously differentiable $x \mapsto \theta(x)$ defined near x_0 , with $\theta(x_0) = \theta^*(\delta)$ and $S_d(x, \theta(x)) = 0$ for every d ; by the characterization above and uniqueness of the maximizer, $\theta(x) = \theta^*(\delta^x)$ for x near x_0 . In particular $x \mapsto \beta^*(\delta^x)$ is differentiable at x_0 , with derivative the treatment coordinate v_β of $v = \theta'(x_0)$. Differentiating $S_d(x, \theta(x)) = 0$ at $x = x_0$ and using Step 2 gives the linearized score: for every direction d ,

$$\sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt}(r'_{gt}d) \{p_{gt} - \mu_{gt}^*(r'_{gt}v)\} = 0.$$

Step 4: solving the linearized score by weighted FWL. Put

$$K = \sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt}\mu_{gt}^* > 0, \quad \psi_{gt} = \frac{q_{gt}\mu_{gt}^*}{K}, \quad \langle f, h \rangle = \sum_{g \in \mathcal{C}} \sum_{t=1}^T \psi_{gt} f_{gt} h_{gt},$$

and define the mean-normalized one-cell source and the nuisance part of $v = (v_a, v_\beta)$ by

$$u_{gt} = \frac{p_{gt}}{\mu_{gt}^*}, \quad \alpha_{gt} = \sum_j X_{gt,j} v_{a,j},$$

so that $r'_{gt}v = \alpha_{gt} + v_\beta D_{gt}$ and α lies in the span of the nuisance columns $X_{\cdot,j}$. Because $\mu_{gt}^* > 0$ and $K\psi_{gt} = q_{gt}\mu_{gt}^*$, each summand of the linearized score factors as

$$q_{gt}(r'_{gt}d) \{p_{gt} - \mu_{gt}^*(r'_{gt}v)\} = K \psi_{gt}(r'_{gt}d) \{u_{gt} - v_\beta D_{gt} - \alpha_{gt}\},$$

so dividing the whole linearized score by $K > 0$ rewrites it as

$$\langle u - v_\beta D - \alpha, (r'_{gt}d)_{g,t} \rangle = 0 \quad \text{for every direction } d.$$

Taking d to be the pure treatment direction gives $\langle u - v_\beta D - \alpha, D \rangle = 0$; taking $d = (\rho, 0)$ for arbitrary ρ gives $\langle u - v_\beta D - \alpha, h \rangle = 0$ for every h in the nuisance span. By [definition 4](#), $D - W$ is the ψ -weighted projection of D on the nuisance span, so W is ψ -orthogonal to that span and $\langle W, D \rangle = \langle W, W \rangle$. Pairing W with $u - v_\beta D - \alpha$ and using both normal equations (the second one at $h = D - W$ and at $h = \alpha$),

$$0 = \langle W, u - v_\beta D - \alpha \rangle = \langle W, u \rangle - v_\beta \langle W, W \rangle.$$

Now $\langle W, W \rangle = \mathcal{E}/K > 0$ by Step 1, and, since p is supported on the single cell (k, s) ,

$$\langle W, u \rangle = \frac{1}{K} \sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt}\mu_{gt}^* W_{gt} \frac{p_{gt}}{\mu_{gt}^*} = \frac{q_{ks} B_{ks} \exp(\delta_{ks}) W_{ks}}{K}.$$

Dividing,

$$v_\beta = \frac{\langle W, u \rangle}{\langle W, W \rangle} = \frac{q_{ks} B_{ks} \exp(\delta_{ks}) W_{ks}}{\mathcal{E}}.$$

Step 5: the derivative formula. Combining Steps 3 and 4,

$$\left. \frac{d}{dx} \beta^*(\delta \text{ with } \delta_{ks} \text{ replaced by } x) \right|_{x=\delta_{ks}} = v_\beta = \frac{q_{ks} B_{ks} \exp(\delta_{ks}) \widetilde{W}_{ks}(\delta)}{\mathcal{E}(\delta)},$$

which is the asserted differentiability at $x = \delta_{ks}$ together with the displayed derivative.

Step 6: the sign equivalences. The derivative is cW_{ks} with

$$c = \frac{q_{ks} B_{ks} \exp(\delta_{ks})}{\mathcal{E}} > 0,$$

because $q_{ks} > 0$, $B_{ks} = \bar{b}_k \exp(\gamma_{s0}) > 0$, $\exp(\delta_{ks}) > 0$, and $\mathcal{E} > 0$ by Step 1. Multiplication by a positive scalar preserves and reflects each of the three order relations, so

$$cW_{ks} < 0 \iff W_{ks} < 0, \quad cW_{ks} = 0 \iff W_{ks} = 0, \quad 0 < cW_{ks} \iff 0 < W_{ks},$$

which, with $W_{ks} = \widetilde{W}_{ks}(\delta)$, are exactly the three asserted equivalences. $\square$

Proof of [theorem 2](#). For a collapsed parameter $\theta \in \mathbb{R}^{|\mathcal{C}|+T}$ and a supported cell (g, t) , write the collapsed linear index as

$$\eta_{gt}(\theta) = r'_{gt}\theta,$$

so that, by [definition 3](#),

$$L(\theta; \delta) = \sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} \{m_{gt}(\delta) \eta_{gt}(\theta) - \exp(\eta_{gt}(\theta))\}, \quad \mu_{gt}^*(\delta) = \exp\{\eta_{gt}(\theta^*(\delta))\}.$$

The map $\theta \mapsto \eta_{gt}(\theta)$ is linear, so $\eta_{gt}(\theta) - \eta_{gt}(\theta') = \eta_{gt}(\theta - \theta')$ for all θ, θ' .

Step 1: the candidate parameter. Let ∞ be the never-treated reference cohort and define $\theta^0 = (a^0, \delta_0)$ coordinatewise by

$$a_{\text{int}}^0 = \log \bar{b}_\infty, \quad a_{\text{coh},c}^0 = \log \bar{b}_c - \log \bar{b}_\infty \quad (c \in \mathcal{C}, c \neq \infty), \quad a_{\text{time},u}^0 = \gamma_{u0} \quad (u \neq 1),$$

with treatment coordinate δ_0 . [Assumption 2](#) carries the normalization $\gamma_{10} = 0$. Hence, for every period t ,

$$\begin{cases} 0, & t = 1, \\ a_{\text{time},t}^0, & t \neq 1, \end{cases} = \gamma_{t0},$$

since for $t = 1$ all period dummies are switched off and $\gamma_{10} = 0$, while for $t \neq 1$ exactly the dummy of period t contributes. Likewise, for every $g \in \mathcal{C}$,

$$\begin{cases} 0, & g = \infty, \\ a_{\text{coh},g}^0, & g \neq \infty, \end{cases} = \begin{cases} 0, & g = \infty, \\ \log \bar{b}_g - \log \bar{b}_\infty, & g \neq \infty, \end{cases}$$

because exactly the dummy of cohort g contributes when $g \neq \infty$, and none when $g = \infty$.

Step 2: the index at θ^0 . Adding the intercept, cohort, period, and treatment contributions in the two cases $g = \infty$ and $g \neq \infty$ gives, for every $g \in \mathcal{C}$ and every period t ,

$$\eta_{gt}(\theta^0) = \log \bar{b}_g + \gamma_{t0} + D_{gt}\delta_0.$$

Step 3: the homogeneous-effect hypothesis in product form. For every $g \in \mathcal{C}$ and every t ,

$$D_{gt}\delta_{gt} = D_{gt}\delta_0 :$$

if $D_{gt} = 1$ the common treated-cell hypothesis gives $\delta_{gt} = \delta_0$, and if $D_{gt} = 0$ both sides vanish.

Step 4: the criterion is exactly specified at θ^0 . Using $B_{gt} = \bar{b}_g \exp(\gamma_{t0})$ together with Steps 2 and 3, the observed collapsed mean satisfies

$$m_{gt}(\delta) = B_{gt} \exp(D_{gt}\delta_{gt}) = \exp\{\log \bar{b}_g + \gamma_{t0} + D_{gt}\delta_0\} = \exp\{\eta_{gt}(\theta^0)\}.$$

In particular $m_{gt}(\delta) > 0$ and

$$\log m_{gt}(\delta) = \eta_{gt}(\theta^0) \quad \text{for every supported cell.}$$

Step 5: the scalar Poisson cell inequality. For every $m > 0$ and every real η ,

$$m\eta - \exp(\eta) \leq m \log m - \exp(\log m),$$

with strict inequality when $\eta \neq \log m$. Apply $1 + x \leq \exp(x)$, strict for $x \neq 0$, at $x = \eta - \log m$, multiply by $m > 0$, and rearrange using $\exp(\eta) = m \exp(\eta - \log m)$.

Step 6: the cell masses are strictly positive. First $T > 0$: if $T = 0$, then for the nonzero collapsed parameter with zero nuisance block and treatment coordinate one the weighted sum $\sum_{g \in C} \sum_{t=1}^T q_{gt}(r'_{gt}a)^2$ is an empty sum, hence zero, contradicting [assumption 5](#). Since $\pi_g \in (0, 1)$, it follows that

$$q_{gt} = \frac{\pi_g}{T} > 0 \quad \text{for every supported cell.}$$

Step 7: θ^0 is a global maximizer. Fix any θ . Applying Step 5 cell by cell with $m = m_{gt}(\delta) > 0$ and $\eta = \eta_{gt}(\theta)$, and using $\log m_{gt}(\delta) = \eta_{gt}(\theta^0)$ from Step 4,

$$m_{gt}(\delta)\eta_{gt}(\theta) - \exp(\eta_{gt}(\theta)) \leq m_{gt}(\delta)\eta_{gt}(\theta^0) - \exp(\eta_{gt}(\theta^0)).$$

Multiplying by $q_{gt} > 0$ and summing over supported cells,

$$L(\theta; \delta) \leq L(\theta^0; \delta).$$

Step 8: the maximizer is unique. Suppose $L(\theta; \delta) = L(\theta^0; \delta)$ and $\theta \neq \theta^0$. Then $\theta - \theta^0 \neq 0$, so [assumption 5](#) gives

$$\sum_{g \in C} \sum_{t=1}^T q_{gt} (\eta_{gt}(\theta - \theta^0))^2 > 0,$$

and by linearity $\eta_{gt}(\theta - \theta^0) = \eta_{gt}(\theta) - \eta_{gt}(\theta^0)$, so some supported cell (g_0, t_0) has

$$\eta_{g_0 t_0}(\theta) \neq \eta_{g_0 t_0}(\theta^0) = \log m_{g_0 t_0}(\delta).$$

At that cell the strict half of Step 5 applies, and all other cells obey the weak inequality of Step 7. Since $q_{g_0 t_0} > 0$, summing gives

$$L(\theta; \delta) < L(\theta^0; \delta),$$

contradicting the assumed equality. Hence $\theta = \theta^0$.

Step 9: identification of the projection. By Steps 7 and 8 the criterion $L(\cdot; \delta)$ has θ^0 as its unique global maximizer, and [definition 3](#) selects $\theta^*(\delta)$ among the global maximizers, so

$$\theta^*(\delta) = \theta^0.$$

Step 10: the two conclusions. The treatment coordinate of θ^0 is δ_0 , so

$$\beta^*(\delta) = \delta_0,$$

and for every $g \in \mathcal{C}$ and every period t , by Step 2,

$$\mu_{gt}^*(\delta) = \exp\{\eta_{gt}(\theta^*(\delta))\} = \exp\{\log \bar{b}_g + \gamma_{t0} + D_{gt}\delta_0\} = B_{gt} \exp(D_{gt}\delta_0).$$

□

Proof of [theorem 3](#). Write $q_{gt} = \pi_g/T$ and $m_{gt} = m_{gt}(\delta) = B_{gt} \exp(D_{gt}\delta_{gt})$ with $B_{gt} = \bar{b}_g \exp(\gamma_{t0})$, so that the quantities of [definition 7](#) satisfy

$$h_{gt} = \pi_g m_{gt} = T q_{gt} m_{gt}, \quad R_g = \sum_{t=1}^T h_{gt}, \quad C_t = \sum_{g \in \mathcal{C}} h_{gt}, \quad M = \sum_{g \in \mathcal{C}} R_g, \quad A = \sum_{g \in \mathcal{C}} \sum_{t=1}^T D_{gt} h_{gt}.$$

A collapsed parameter is written $\theta = (a, \beta)$ with nuisance block a and treatment coordinate β , and $r'_{gt}\theta = \sum_j X_{gt,j} a_j + D_{gt}\beta$. Finally set

$$B_{gt}^{obs} = \frac{m_{gt} m_{\infty t}}{m_{\infty 1}}, \quad H = \{(g, t) : g \in \mathcal{C}, D_{gt} = 1\}, \quad Z = \sum_{(g,t) \in H} q_{gt} B_{gt}^{obs},$$

$$\tau_{gt} = \frac{m_{gt}}{B_{gt}^{obs}} - 1, \quad \omega_{gt} = \frac{q_{gt} B_{gt}^{obs}}{Z}, \quad PTT = \sum_{(g,t) \in H} \omega_{gt} \tau_{gt}.$$

Step 1: positivity of the primitive margins. Since $T \geq 4$, in particular $T > 0$, and $\infty \in \mathcal{C}$, so $\mathcal{C}$ is nonempty. Every $h_{gt} = \pi_g \bar{b}_g \exp(\gamma_{t0}) \exp(D_{gt}\delta_{gt})$ is a product of strictly positive factors, hence

$$h_{gt} > 0, \quad R_g > 0, \quad C_t > 0, \quad M > 0.$$

Also, summing h_{gt} in the two possible orders,

$$\sum_{t=1}^T C_t = M = \sum_{g \in \mathcal{C}} R_g.$$

Step 2: the zero-treatment row-column fit. Define the collapsed parameter $\theta^0 = (a^0, 0)$, whose treatment coordinate is zero and whose nuisance levels are

$$a_{\text{int}}^0 = \log \frac{R_\infty}{\pi_\infty} + \log C_1 - \log M, \quad a_{\text{coh},g}^0 = \log \frac{R_g}{\pi_g} - \log \frac{R_\infty}{\pi_\infty} \quad (g \neq \infty), \quad a_{\text{time},t}^0 = \log C_t - \log C_1 \quad (t \neq 1).$$

Adding the intercept, cohort, and period contributions — the cohort dummy being absent for $g = \infty$ and the period dummy for $t = 1$ — gives, for every supported cell,

$$r'_{gt}\theta^0 = \log \frac{R_g}{\pi_g} + \log C_t - \log M, \quad \text{hence} \quad \exp(r'_{gt}\theta^0) = \frac{R_g C_t}{\pi_g M},$$

the exponentiation being legitimate because R_g/π_g , C_t , and M are strictly positive by Step 1.

Step 3: the conditional residuals and their margins. Define the mass-weighted residual of θ^0 in cell (g, t) ,

$$e_{gt} = q_{gt} \left\{ m_{gt} - \exp(r'_{gt}\theta^0) \right\} = \frac{\pi_g}{T} \left\{ m_{gt} - \frac{R_g C_t}{\pi_g M} \right\} = \frac{1}{T} \left\{ h_{gt} - \frac{R_g C_t}{M} \right\}.$$

Its row and column sums vanish: using $\sum_t C_t = M$ and $\sum_{g \in \mathcal{C}} R_g = M$ from Step 1,

$$\begin{aligned} \sum_{t=1}^T e_{gt} &= \frac{1}{T} \left\{ R_g - \frac{R_g}{M} \sum_{t=1}^T C_t \right\} = 0 \quad (g \in \mathcal{C}), \\ \sum_{g \in \mathcal{C}} e_{gt} &= \frac{1}{T} \left\{ C_t - \frac{C_t}{M} \sum_{g \in \mathcal{C}} R_g \right\} = 0 \quad (t = 1, \dots, T). \end{aligned}$$

Step 4: the scores at θ^0 . Every collapsed nuisance regressor $X_{\cdot,j}$ is either the constant 1, the indicator of one cohort $g_0 \neq \infty$, or the indicator of one period $t_0 \neq 1$. The corresponding score at θ^0 is therefore $\sum_{g,t} e_{gt}$, $\sum_t e_{g_0 t}$, or $\sum_g e_{g t_0}$, and all three vanish by Step 3:

$$\sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} X_{gt,j} \{m_{gt} - \exp(r'_{gt} \theta^0)\} = 0 \quad \text{for every nuisance coordinate } j.$$

The treatment score at θ^0 is, by Step 3 and the definitions of A and Φ ,

$$S = \sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} D_{gt} \{m_{gt} - \exp(r'_{gt} \theta^0)\} = \frac{1}{T} \left\{ A - \frac{1}{M} \sum_{g \in \mathcal{C}} \sum_{t=1}^T D_{gt} R_g C_t \right\} = \frac{\Phi}{TM},$$

and $TM > 0$ by Step 1.

Step 5: the pseudo-true coefficient has the sign of S . By [lemma 2](#), $\theta^*(\delta)$ is the unique global maximizer of $L(\cdot; \delta)$ and every directional score vanishes there:

$$\sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} (r'_{gt} d) \{m_{gt} - \mu_{gt}^*(\delta)\} = 0 \quad \text{for every direction } d.$$

Write $v = \theta^*(\delta) - \theta^0$, whose treatment coordinate is $v_\beta = \beta^*(\delta)$ because θ^0 has treatment coordinate zero. Decomposing an arbitrary direction $d = (d_a, d_\beta)$ into its nuisance and treatment parts and using Step 4, the score at θ^0 in direction d is

$$\sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} (r'_{gt} d) \{m_{gt} - \exp(r'_{gt} \theta^0)\} = d_\beta S.$$

Suppose first $v \neq 0$. Subtracting the vanishing score at $\theta^*(\delta)$ in direction v from the score at θ^0 in direction v ,

$$\beta^*(\delta) S = \sum_{g \in \mathcal{C}} \sum_{t=1}^T q_{gt} (r'_{gt} v) \{\exp(r'_{gt} \theta^*(\delta)) - \exp(r'_{gt} \theta^0)\}.$$

Each summand is nonnegative: $q_{gt} > 0$, and since $r'_{gt} \theta^*(\delta) - r'_{gt} \theta^0 = r'_{gt} v$ and the exponential is increasing, $r'_{gt} v$ and $\exp(r'_{gt} \theta^*(\delta)) - \exp(r'_{gt} \theta^0)$ have the same sign. Moreover $v \neq 0$ and the injectivity of $\theta \mapsto (r'_{gt} \theta)_{(g,t)}$ established from [assumption 5](#) in [lemma 2](#) give a supported cell with $r'_{gt} v \neq 0$, where the summand is strictly positive by strict monotonicity. Hence

$$\beta^*(\delta) S > 0 \quad \text{whenever } \theta^*(\delta) \neq \theta^0.$$

If instead $v = 0$, then $\beta^*(\delta) = 0$, and applying the vanishing score at $\theta^*(\delta) = \theta^0$ to the pure treatment direction gives $S = 0$. Conversely, if $S = 0$ then by the previous display every score at θ^0 vanishes in every direction, so θ^0 is a global maximizer of $L(\cdot; \delta)$ — for a Poisson criterion with nonnegative weights the tangent-line bound $\exp(\eta') \geq \exp(\eta)\{1 + (\eta' - \eta)\}$ makes vanishing scores sufficient for global maximality — and uniqueness forces $\theta^*(\delta) = \theta^0$, i.e. $\beta^*(\delta) = 0$. Collecting the two cases: $\beta^*(\delta) = 0 \iff S = 0$, and when both are nonzero their product is positive, so they have the same sign. Since $S = \Phi/(TM)$ with $TM > 0$, the sign of S is the sign of Φ , and therefore

$$\beta^*(\delta) < 0 \iff \Phi < 0, \quad \beta^*(\delta) = 0 \iff \Phi = 0, \quad 0 < \beta^*(\delta) \iff 0 < \Phi.$$

Step 6: the observed baseline proxy reproduces the untreated mean. For every supported cohort g the first-period treatment indicator is zero: if $g = \infty$ this is the never-treated path, and if g is finite then $D_{g1} = 1$ would force the adoption date to be the first period, which the support excludes. Also $D_{\infty t} = 0$ for every t , and [assumption 2](#) gives $\gamma_{10} = 0$. Hence, for every $g \in \mathcal{C}$ and every t ,

$$B_{gt}^{obs} = \frac{m_{g1}m_{\infty t}}{m_{\infty 1}} = \frac{\bar{b}_g e^{\gamma_{10}} \cdot \bar{b}_{\infty} e^{\gamma_{t0}}}{\bar{b}_{\infty} e^{\gamma_{10}}} = \bar{b}_g e^{\gamma_{t0}} = B_{gt},$$

the cancellation being legitimate because $m_{\infty 1} = \bar{b}_{\infty} > 0$.

Step 7: H is nonempty and $Z > 0$. [Assumption 6](#) supplies a supported finite cohort adopting in the second period; in that period its treatment indicator equals one, so H is nonempty. For every $(g, t) \in H$ we have $q_{gt} > 0$ and, by Step 6, $B_{gt}^{obs} = B_{gt} > 0$. Hence every summand of Z is positive and at least one exists:

$$Z = \sum_{(g,t) \in H} q_{gt} B_{gt}^{obs} > 0.$$

Step 8: membership in $\mathcal{R}_T$ is equivalent to $\Phi < 0$. If the induced primitive tuple lies in $\mathcal{R}_T$, then the last clause of [definition 5](#) asserts negativity of its treatment coordinate; that coordinate is the treatment coordinate of the maximizer of the criterion written in the restricted primitive coordinates, which is the same function of θ as $L(\theta; \delta)$, so it equals $\beta^*(\delta)$, and Step 5 gives $\Phi < 0$. Conversely suppose $\Phi < 0$. The horizon, support, and frontier-scope clauses of [definition 5](#) are hypotheses of the present theorem. The simplex clause holds because the finite-array shares sum to one for every $N \geq |\mathcal{C}|$,

$$\sum_{g \in \mathcal{C}} \frac{n_{gN}}{N} = 1,$$

every unit label lying in $\mathcal{C}$, and passing to the limit with [assumption 1](#) gives $\sum_{g \in \mathcal{C}} \pi_g = 1$. The untreated-mean clause is witnessed by $g \mapsto \bar{b}_g$, since the primitive untreated coordinate in cell (g, t) is $B_{gt} = \bar{b}_g e^{\gamma_{t0}}$. The positive-effect clause is [assumption 7](#), and the rank clause is [assumption 5](#) after rewriting the sum over supported cells as the double sum over $g \in \mathcal{C}$ and t . Finally Step 5 gives $\beta^*(\delta) < 0$, which is the tuple's treatment coordinate. Hence the tuple lies in $\mathcal{R}_T$.

Step 9: the four-cohort polynomial. Let $x > 1$, $y > 1$, and consider the four-period support $\{2, 3, 4, \infty\}$ with $T = 4$, equal shares $\pi_g = 1/4$, unit limiting baselines, zero untreated time effects (so $B_{gt} = 1$), multiplier x on every treated cell except $(2, 4)$, and multiplier y at $(2, 4)$; that is, $\exp(\delta_{gt}) = x$ on treated cells other than $(2, 4)$ and $\exp(\delta_{2,4}) = y$. With treatment

absorbing from the adoption date on, so that $D_{gt} = 1$ exactly when $g \leq t$ and $D_{\infty t} = 0$, the array $h_{gt} = \pi_g B_{gt} \exp(D_{gt} \delta_{gt})$ is

$$4(h_{gt}) = \begin{pmatrix} 1 & x & x & y \\ 1 & 1 & x & x \\ 1 & 1 & 1 & x \\ 1 & 1 & 1 & 1 \end{pmatrix},$$

with rows $g = 2, 3, 4, \infty$ and columns $t = 1, 2, 3, 4$. Summing rows and columns,

$$4R_2 = 1 + 2x + y, \quad 4R_3 = 2 + 2x, \quad 4R_4 = 3 + x, \quad 4R_\infty = 4,$$

$$4C_1 = 4, \quad 4C_2 = x + 3, \quad 4C_3 = 2x + 2, \quad 4C_4 = 2x + y + 1,$$

$$4M = 10 + 5x + y, \quad 4A = 5x + y.$$

The treated cells of row $g = 2$ are $t = 2, 3, 4$, of row $g = 3$ are $t = 3, 4$, and of row $g = 4$ is $t = 4$, so

$$16 \sum_{g,t} D_{gt} R_g C_t = (1 + 2x + y)(5x + y + 6) + (2 + 2x)(4x + y + 3) + (3 + x)(2x + y + 1),$$

which expands to $20x^2 + 10xy + y^2 + 38x + 12y + 15$, while

$$16 MA = (10 + 5x + y)(5x + y) = 25x^2 + 10xy + y^2 + 50x + 10y.$$

Subtracting,

$$\Phi = MA - \sum_{g,t} D_{gt} R_g C_t = \frac{5x^2 + 12x - 2y - 15}{16}.$$

Step 10: the numerical identity. With $5(101/100)^2 = 51005/10000$ and $12(101/100) = 1212/100$,

$$\frac{5(101/100)^2 + 12(101/100) - 15}{2} = \frac{4441}{4000}.$$

Step 11: the value at the witness. For W_4 of [definition 6](#) the treated multipliers are $\exp(\log(101/100)) = 101/100$ off cell $(2, 4)$ and $\exp(\log 4) = 4$ at $(2, 4)$, so Step 9 applies with $x = 101/100$ and $y = 4$:

$$\Phi = \frac{5(101/100)^2 + 12(101/100) - 2 \cdot 4 - 15}{16} = -\frac{11559}{32000}.$$

Step 12: the counterfactual-share objects on treated cells. Let $(g, t) \in H$, so $D_{gt} = 1$. Step 6 gives $B_{gt}^{obs} = B_{gt}$, and $B_{gt} > 0$, so

$$\tau_{gt} = \frac{m_{gt}}{B_{gt}^{obs}} - 1 = \frac{B_{gt} \exp(\delta_{gt})}{B_{gt}} - 1 = \exp(\delta_{gt}) - 1,$$

and, using $q_{gt} > 0$, $B_{gt}^{obs} > 0$, and $Z > 0$ from Step 7,

$$\omega_{gt} = \frac{q_{gt} B_{gt}^{obs}}{Z} > 0.$$

Step 13: the weights sum to one. By the definition of Z and $Z > 0$,

$$\sum_{(g,t) \in H} \omega_{gt} = \frac{1}{Z} \sum_{(g,t) \in H} q_{gt} B_{gt}^{obs} = \frac{Z}{Z} = 1.$$

Step 14: the ratio form of PTT. Expanding the definition and using $B_{gt}^{obs} > 0$ to cancel it inside each term,

$$PTT = \sum_{(g,t) \in H} \frac{q_{gt} B_{gt}^{obs}}{Z} \left(\frac{m_{gt}}{B_{gt}^{obs}} - 1 \right) = \frac{\sum_{(g,t) \in H} q_{gt} m_{gt}}{Z} - \frac{\sum_{(g,t) \in H} q_{gt} B_{gt}^{obs}}{Z},$$

and the second fraction equals 1 by Step 13, so

$$PTT = \frac{\sum_{(g,t) \in H} q_{gt} m_{gt}}{\sum_{(g,t) \in H} q_{gt} B_{gt}^{obs}} - 1.$$

Step 15: the sign contrast. Suppose $\Phi < 0$. Step 5 gives $\beta^*(\delta) < 0$. For PTT , each treated cell has $\omega_{gt} > 0$ by Step 12, and [assumption 7](#) gives $\delta_{gt} > 0$, hence $\exp(\delta_{gt}) > 1$ and $\tau_{gt} > 0$. The sum defining PTT thus has no negative term, and by Step 7 it has at least one term, which is strictly positive. Therefore $PTT > 0$, and

$$\Phi < 0 \implies \beta^*(\delta) < 0 < PTT.$$

□

Proof of Four-Cohort Sign Reversal. 1. The defining conditions of the region in [definition 5](#) are checked directly for the displayed primitive. The horizon condition is $4 \leq 4$. The support contains the never-treated cohort and the finite cohorts 2, 3, 4, all dated after period 1, and these four cohorts give the required four-cohort frontier scope. The shares sum to

$$\sum_{g \in \mathcal{C}} \pi_g = 4 \cdot \frac{1}{4} = 1,$$

and, since $\bar{b}_g = 1$ and $\gamma_{t0} = 0$, the untreated mean representation is $1 = 1 \cdot \exp(0)$ in every supported cell.

For rank, let a be a collapsed parameter whose weighted quadratic design norm is zero. Each summand is nonnegative and each supported cell mass is nonzero, so every collapsed index $r'_{gt} a$ is zero. Evaluating these zero equations first at the never-treated cohort in period 1 gives the intercept equal to zero; evaluating finite cohorts in period 1 gives all cohort dummy coordinates equal to zero; evaluating the never-treated cohort across periods gives all time dummy coordinates equal to zero; and evaluating the cohort 2, period 2, treated cell then gives the treatment coordinate equal to zero. Hence $a = 0$, proving the collapsed design rank condition.

The sign condition is obtained from the primitive sign index Φ in [definition 7](#). In the multiplier subfamily with common treated multiplier x off cell (2, 4) and multiplier y at (2, 4),

$$\Phi = \frac{5x^2 + 12x - 2y - 15}{16}.$$

Substituting $x = 101/100$ and $y = 4$ gives

$$\Phi = -\frac{11559}{32000} < 0.$$

The sign characterization of the frontier index gives $\beta^*(\delta) < 0$, and the restricted primitive has the same primitive treatment coordinate. Together with the strict positivity of the treated effects verified in the next step, all clauses of [definition 5](#) hold.

2. If $D_{gt} = 1$, then the specified effect is either $\log 4$ or $\log(101/100)$. Both are strictly positive because $4 > 1$ and $101/100 > 1$.
3. For the designated cell $(2, 4)$, treatment is on and the effect is $\delta_{2,4} = \log 4$. For every other treated supported cell, the effect is $\log(101/100)$. Since $101/100 < 4$ and the logarithm is strictly increasing on positive numbers,

$$\log(101/100) < \log 4 = \delta_{2,4}.$$

4. At the zero-effect vector, the fitted mean is 1 in every supported cell by [theorem 2](#). Hence the mean weights in [definition 4](#) are all $1/16$. The displayed candidate residual table has row sums and column sums all equal to zero. Moreover, treatment minus this table equals

$$-\frac{3}{8} + c_g + s_t,$$

with

$$(c_2, c_3, c_4, c_\infty) = \left(\frac{3}{4}, \frac{1}{2}, \frac{1}{4}, 0\right), \quad (s_1, s_2, s_3, s_4) = \left(0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right),$$

so treatment minus the table lies in the nuisance fixed-effect space. The zero row and column sums make the table orthogonal, under the constant weights $1/16$, to every nuisance regressor and therefore to the whole nuisance space. Thus the weighted projection of treatment onto the nuisance space is treatment minus the table, and the residual is the table. This gives exactly

$$\widetilde{W}_{gt} = \frac{1}{8} \begin{pmatrix} -3 & 3 & 1 & -1 \\ -1 & -3 & 3 & 1 \\ 1 & -1 & -3 & 3 \\ 3 & 1 & -1 & -3 \end{pmatrix}_{gt}.$$

5. Evaluating the residual table at the designated cell gives

$$\widetilde{W}_{2,4}(0) = -\frac{1}{8}.$$

6. The residual energy at zero effects is

$$\sum_{g,t} q_{gt} \mu_{gt}^*(0) \widetilde{W}_{gt}(0)^2 = \frac{1}{16} \sum_{g,t} \widetilde{W}_{gt}(0)^2 = \frac{1}{16} \cdot \frac{80}{64} = \frac{5}{64}.$$

[theorem 1](#), applied to the treated cell $(2, 4)$, gives

$$\left. \frac{d}{dx} \beta^*(\delta_{2,4} = x, \delta_{gt} = 0 \text{ otherwise}) \right|_{x=0} = \frac{q_{2,4} B_{2,4} \exp(0) \widetilde{W}_{2,4}(0)}{5/64}.$$

Here $q_{2,4} = 1/16$, $B_{2,4} = 1$, and $\widetilde{W}_{2,4}(0) = -1/8$. Therefore the derivative is

$$\frac{(1/16) \cdot 1 \cdot 1 \cdot (-1/8)}{5/64} = -\frac{1}{10}.$$

7. By continuity of the weighted FWL residual in the effect vector, $\widetilde{W}_{2,4}(\delta)$ is continuous at the zero-effect vector. Since $\widetilde{W}_{2,4}(0) = -1/8 < 0$, the inverse image of $(-\infty, 0)$ is a neighborhood of zero. Choose a finite coordinate neighborhood contained in that inverse image, choose a positive radius for each coordinate neighborhood, and let ε be the minimum of those radii, or 1 if the finite coordinate set is empty.

Now take any δ' such that $|\delta'_{gt}| < \varepsilon$ on every supported cell and $\delta'_{gt} > 0$ on every treated supported cell. Define the supported truncation δ^s by $\delta^s_{gt} = \delta'_{gt}$ on $g \in \mathcal{C}$ and $\delta^s_{gt} = 0$ outside the support. The choice of ε puts δ^s in the coordinate neighborhood, hence $\widetilde{W}_{2,4}(\delta^s) < 0$. Changing effects outside supported cohorts leaves the limiting criterion, fitted means, and weighted FWL residuals unchanged, so $\widetilde{W}_{2,4}(\delta') = \widetilde{W}_{2,4}(\delta^s) < 0$. Applying [theorem 1](#) again at the treated cell $(2, 4)$, the derivative of $x \mapsto \beta^*$ with the $(2, 4)$ coordinate of δ' replaced by x , evaluated at $x = \delta'_{2,4}$, has the same sign as $\widetilde{W}_{2,4}(\delta')$, because the denominator and multiplicative prefactor are strictly positive. Thus this derivative is strictly negative for every such strictly positive δ' .

8. The seven conclusions above are assembled in the order stated: region membership, positivity of treated effects, uniqueness of the largest treated effect at $(2, 4)$, the full weighted FWL residual table, the late-cell residual $-1/8$, the exact late-cell derivative at zero, and the local negative derivative statement under strict positivity of treated supported effects. $\square$

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